

37495 Statistical Design and Models for Evaluation Studies

Assessment Task 2: Assignment

This assessment task is marked out of 50

Please submit in PDF format with file name

37495_Assignment_LastName_FirstName

Include all R code and any output you refer to in your report

Due 23:59PM Monday 20 April 2026

Use of R packages: Students are expected to use the R packages and functions introduced in the workshops for this assignment.

If you choose to use alternative R packages or functions, you must:

- demonstrate that the results are equivalent (up to simulation or numerical variation),
or
- clearly explain the reasons for any differences in results (e.g. different default settings, algorithms, or assumptions).

Answers that use alternative packages without justification or explanation may not receive full marks.

Question 1 [10 marks]

Three technicians in a chemical plant are given the task of investigating the yield of a chemical process by using one of five chemicals as inputs. Three responses are to be observed for each chemical input. Design an appropriate experiment suitable for this purpose. Make sure you identify the response variable, the experimental factor, the blocking factor, the measurement units and describe how treatments should be allocated.

Question 2 [30 marks]

A manufacturer of roofing insulation wanted to test their product for impact strength as part of its quality control procedure. A completely randomised design experiment was conducted where twenty samples each from five daily production runs were randomly selected and tested, with impact strength measured and recorded. The data is presented in the table below.

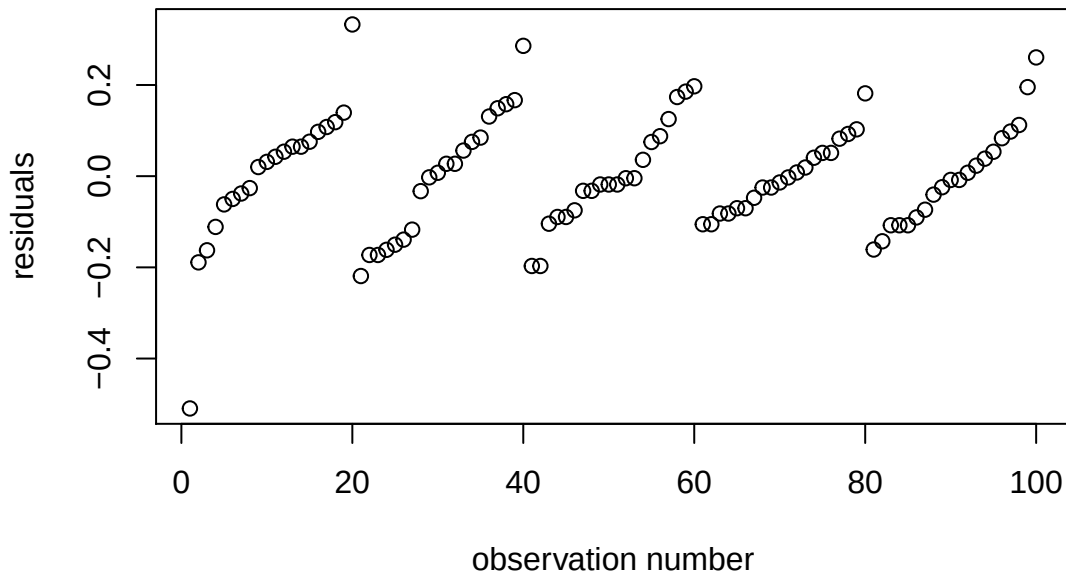
<i>run</i>	<i>strength</i>									
1	0.46	0.67	0.69	0.73	0.77	0.78	0.79	0.80	0.84	0.85
	0.86	0.87	0.88	0.88	0.89	0.91	0.92	0.93	0.95	1.15
2	0.80	0.84	0.84	0.85	0.86	0.87	0.89	0.97	1.00	1.01
	1.03	1.03	1.06	1.08	1.09	1.14	1.16	1.17	1.18	1.32
3	0.52	0.52	0.58	0.59	0.59	0.60	0.63	0.63	0.64	0.64
	0.64	0.65	0.65	0.68	0.71	0.72	0.75	0.79	0.80	0.81
4	0.79	0.79	0.81	0.81	0.82	0.82	0.84	0.86	0.86	0.87
	0.88	0.89	0.90	0.92	0.93	0.93	0.96	0.97	0.98	1.06
5	0.44	0.45	0.47	0.47	0.47	0.48	0.49	0.51	0.52	0.53
	0.53	0.54	0.55	0.56	0.57	0.59	0.60	0.61	0.67	0.72

- Perform a hypothesis test to determine if mean impact strength of insulation from production run 1 is greater than 0.78 [4 marks].
- Create a “bootstrapped” distribution of the sample mean of impact strength for production run 1 by re-sampling this sample data with replacement 100000 times, computing the mean for each re-sample and plotting these in a histogram with 100 bins. Over the top of this histogram fit a normal density curve to the data and assess whether the bootstrapped sample mean is approximately normally-distributed using

the plot produced and statistical test (based on a random sample of 50 observations). [4 marks].

- (c) Fit an ANOVA model and determine whether the mean impact strength of the insulation varies with production run, using 0.05 significance level [4 marks].
- (d) Perform appropriate hypothesis tests to determine if the residuals from the model in (c) are normally-distributed and have constant variance, using 0.05 significance level [4 marks].
- (e) Apply the optimal Box-Cox transform to the response variable $y_{ij} = strength_{ij}$, making sure to report the value of λ applied [2 marks].
- (f) Using the Box-Cox transformed data, fit an ANOVA model and perform appropriate hypothesis tests to determine if the residuals from this model are normally-distributed and have constant variance, using 0.05 significance level [4 marks].
- (g) Using the fitted model from (f) and Scheffe's method to control family-wise error rate to 0.05, perform hypothesis tests to determine if there are significant differences in mean impact strength for
 - i. production run 1 versus the average of production runs 2-5
 - ii. production run 1 versus production run 3
 - iii. the average of production runs 1, 2, 4 versus the average of production runs 3, 5using 0.05 significance level [4 marks].
- (h) The Durbin-Watson test of autocorrelation applied to the residuals from the fitted models results in the null hypothesis of no autocorrelation being rejected. A plot of the residuals against observation number is below.

Plot of residuals from Box–Cox model



Explain why such clear patterns appear in this case [**2 marks**].

- (i) Save Cook's Distances into your data frame by passing the fitted “aov” object from (f) into the R function “cooks.distance”. Using R, create a filtered data set consisting of the Box-Cox transformed response data and the production run data by removing observations with Cook's D greater than 0.05. Print out the first 40 records of this filtered data set [**2 marks**].

Question 3 [10 marks]

Consider a completely randomised design with statistical model

$$y_{ij} = \mu + \tau_i + e_{ij}, \quad i \in \{1, \dots, a\}, \quad j \in \{1, \dots, n\},$$

with

$$\sum_{i=1}^a \tau_i = 0$$

where $e_{ij} \sim N(0, \sigma^2)$ and independent.

- (a) Showing and justifying each step, derive the normal equations and obtain the least squares estimates

$$\hat{\mu} = \bar{y}_{..},$$

$$\hat{\tau}_i = \bar{y}_i - \bar{y}_{..}$$

where

$$\bar{y}_{..} = \frac{1}{an} \sum_{i=1}^a \sum_{j=1}^n y_{ij},$$

$$\bar{y}_{i.} = \frac{1}{n} \sum_{j=1}^n y_{ij}.$$

(b) Showing and justifying each step, find $\text{var}(\hat{e}_{ij})$ where $\hat{e}_{ij}$ is the residual

$$\hat{e}_{ij} = y_{ij} - \hat{\mu} - \hat{\tau}_i.$$